

Name: _____

Date: _____

UNIT 6 · FUNCTIONS · GUIDED NOTES

Tables & Systems of Equations

Fill in the blanks. Practice problems at the end.

1 Writing Linear Equations from a Table

Every linear equation can be written as $y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$, where **m** is the $\underline{\hspace{1cm}}$ (rate of change) and **b** is the $\underline{\hspace{1cm}}$ (the y-value when $x = 0$).

$$m = (y_2 - y_1) \div (x_2 - x_1) = \text{change in } y \div \text{change in } x$$

Example 1 · Table includes $x = 0$

x	y
0	3
1	7
2	11
3	15

Step 1. From (0, 3) to (1, 7):change in $y = \underline{\hspace{1cm}}$, change in $x = \underline{\hspace{1cm}}$ slope $m = \underline{\hspace{1cm}}$ **Step 2.** When $x = 0$, $y = \underline{\hspace{1cm}}$. So $b = \underline{\hspace{1cm}}$.**Step 3. Equation:** $y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$ **Example 2 · No $x = 0$ row — work backwards for b**

x	y
2	5
4	11
6	17
8	23

Step 1. From (2, 5) to (4, 11):change in $y = \underline{\hspace{1cm}}$, change in $x = \underline{\hspace{1cm}}$. $m = \underline{\hspace{1cm}}$.**Step 2.** Use $b = y - m \cdot x$. Pick (2, 5): $b = \underline{\hspace{1cm}} - (\underline{\hspace{1cm}})(\underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$ **Step 3. Equation:** $y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$ **SHORTCUT FOR FINDING B**Once you know m , pick **any** row (x, y) . Then: $b = y - m \cdot x$

2 Solving Systems of Equations by Graphing

A **system of equations** is or more equations that use the same . In 8th grade, we work with systems of **two** linear equations in x and y .

Graphing Method

Graph each line on the same coordinate plane. The solution is the point.

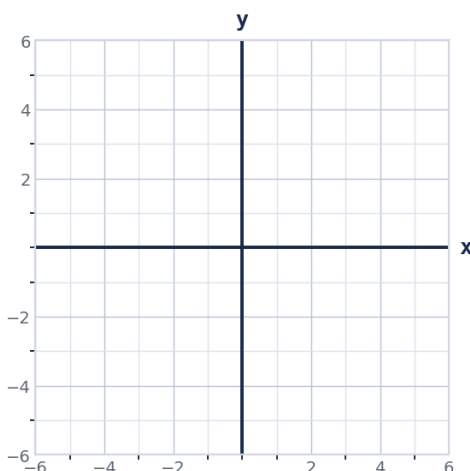
STEPS

1. Graph the first line (using slope and y-intercept).
2. Graph the second line on the same plane.
3. Find where they . That ordered pair is the solution.
4. Check by plugging the point into both equations.

Example · Graph and solve

EQ 1 $y = x + 2$

EQ 2 $y = -x + 4$



EQ 1: $y = x + 2$

slope = , y-int =

EQ 2: $y = -x + 4$

slope = , y-int =

Solution (intersection):

(,)

Check:

EQ 1: = + 2 ✓

EQ 2: = - + 4 ✓

Three Possible Outcomes

Case	What You See	How Many Solutions?
1	Lines cross at one point	<u> </u> solution
2	Lines are <u> </u> (never cross)	<u> </u> solutions
3	Both equations graph the same line	<u> </u> solutions

3 What Does "Solution" Actually Mean?

A **solution** to a system of equations is an ordered pair (x, y) that makes _____ equations _____ at the same time.

TWO WAYS TO SAY IT

- **Graphically:** the solution is the _____ where the lines cross.
- **Algebraically:** the solution is the (x, y) that, when plugged in, makes _____ equations true.

Testing Whether a Point Is a Solution

Plug x and y into _____ equation. If the point makes _____ equations true, it is a solution. If it fails even one, it is **not** a solution.

Example · Is $(2, 5)$ a solution?

EQ 1 $y = 2x + 1$

EQ 2 $y = 3x - 1$

1 Test EQ 1 with $x = 2, y = 5$: _____ $= 2(\text{ }) + 1 = \text{ }.$ True? _____

2 Test EQ 2 with $x = 2, y = 5$: _____ $= 3(\text{ }) - 1 = \text{ }.$ True? _____

3 Both true, so $(2, 5)$ _____ a solution.

TAKEAWAY

You **cannot** tell just by looking at a point. Always test _____ equations.

4 Solving Systems by Substitution

Graphing shows the solution but is slow and hard to read when answers aren't whole numbers. Substitution is algebraic — it always gives an exact answer.

SUBSTITUTION PROCEDURE

1. One equation should be solved for a variable (usually $y = \dots$).
2. **Substitute** the expression into the _____ equation.
3. Solve that equation for _____.
4. Plug that value back in to find the _____ variable.
5. Write the solution as an _____ pair (x, y) and check.

Example 1 · Both equations in $y =$ form

EQ 1 $y = 2x + 3$

EQ 2 $y = -x + 6$

1 Both equations equal y , so they equal each other: _____ $= -x + 6$

2 Solve for x . Add x : $3x + 3 = 6$. Subtract 3: $3x = \text{ }.$ Divide: $x = \text{ }.$

3 Back-substitute into EQ 1: $y = 2(\text{ }) + 3 = \text{ }.$

4 **Solution:** $(\text{ }, \text{ })$

Example 2 · One equation in $y =$ form, one not

EQ 1 $y = 3x - 1$

EQ 2 $2x + y = 9$

- 1 EQ 1 is already solved for y : $y = \underline{\hspace{2cm}}$
- 2 Substitute ($\underline{\hspace{2cm}}$) for y in EQ 2: $2x + (\underline{\hspace{2cm}}) = 9$
- 3 Combine like terms: $\underline{\hspace{2cm}}x - 1 = 9$. Add 1: $5x = \underline{\hspace{2cm}}$. Divide: $x = \underline{\hspace{2cm}}$.
- 4 Back-substitute: $y = 3(\underline{\hspace{2cm}}) - 1 = \underline{\hspace{2cm}}$.
- 5 **Solution:** ($\underline{\hspace{2cm}}$, $\underline{\hspace{2cm}}$)

PRACTICE PROBLEMS

Show your work. Check your answers.

1. Write the linear equation for this table.

x	y
0	-4
1	-1
2	2
3	5

slope $m = \underline{\hspace{2cm}}$
 y-intercept $b = \underline{\hspace{2cm}}$

Equation: $y = \underline{\hspace{2cm}}x + \underline{\hspace{2cm}}$ **2. Write the linear equation for this table. (no $x = 0$ row)**

x	y
2	7
4	13
6	19
8	25

slope $m = \underline{\hspace{2cm}}$
 Use $b = y - m \cdot x$ with ($\underline{\hspace{2cm}}$, $\underline{\hspace{2cm}}$): $b = \underline{\hspace{2cm}} - (\underline{\hspace{2cm}})(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}}$

Equation: $y = \underline{\hspace{2cm}}x + \underline{\hspace{2cm}}$ **3. Is (3, 5) a solution to this system? Show your work. Circle YES or NO.**

EQ 1 $y = 2x - 1$

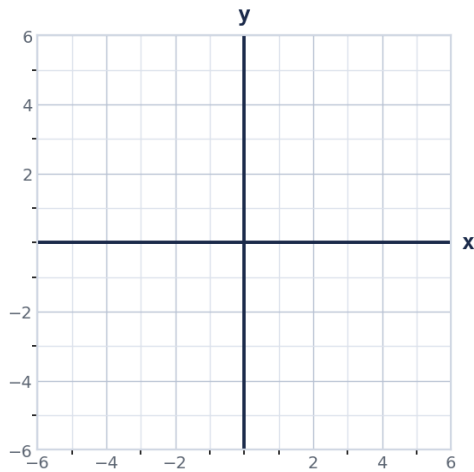
EQ 2 $y = -x + 8$

EQ 1 test: $\underline{\hspace{2cm}} = 2(\underline{\hspace{2cm}}) - 1 = \underline{\hspace{2cm}}$ True or False? $\underline{\hspace{2cm}}$ EQ 2 test: $\underline{\hspace{2cm}} = -(\underline{\hspace{2cm}}) + 8 = \underline{\hspace{2cm}}$ True or False? $\underline{\hspace{2cm}}$ Final answer (circle): **YES** / **NO**

4. Solve this system by graphing on the grid below.

EQ 1 $y = x + 1$

EQ 2 $y = -x + 5$



EQ 1 slope = ____, y-int = ____

EQ 2 slope = ____, y-int = ____

Solution: (____, ____)**Check:**

EQ 1: ____ = ____ + 1

EQ 2: ____ = -____ + 5

5. Solve by substitution.

EQ 1 $y = x + 3$

EQ 2 $y = 2x - 1$

Set equal: ____ = $2x - 1$ Solve for x: $x =$ ____Find y: $y =$ ____ + 3 = ____**Solution:** (____, ____)**6. Solve by substitution. (One equation needs a quick substitution.)**

EQ 1 $y = 3x + 2$

EQ 2 $x + y = 10$

Substitute (____) for y in the second equation: $x +$ (____) = 10Combine: ____ $x +$ ____ = 10Solve for x: $x =$ ____Find y: $y = 3$ (____) + 2 = ____**Solution:** (____, ____)